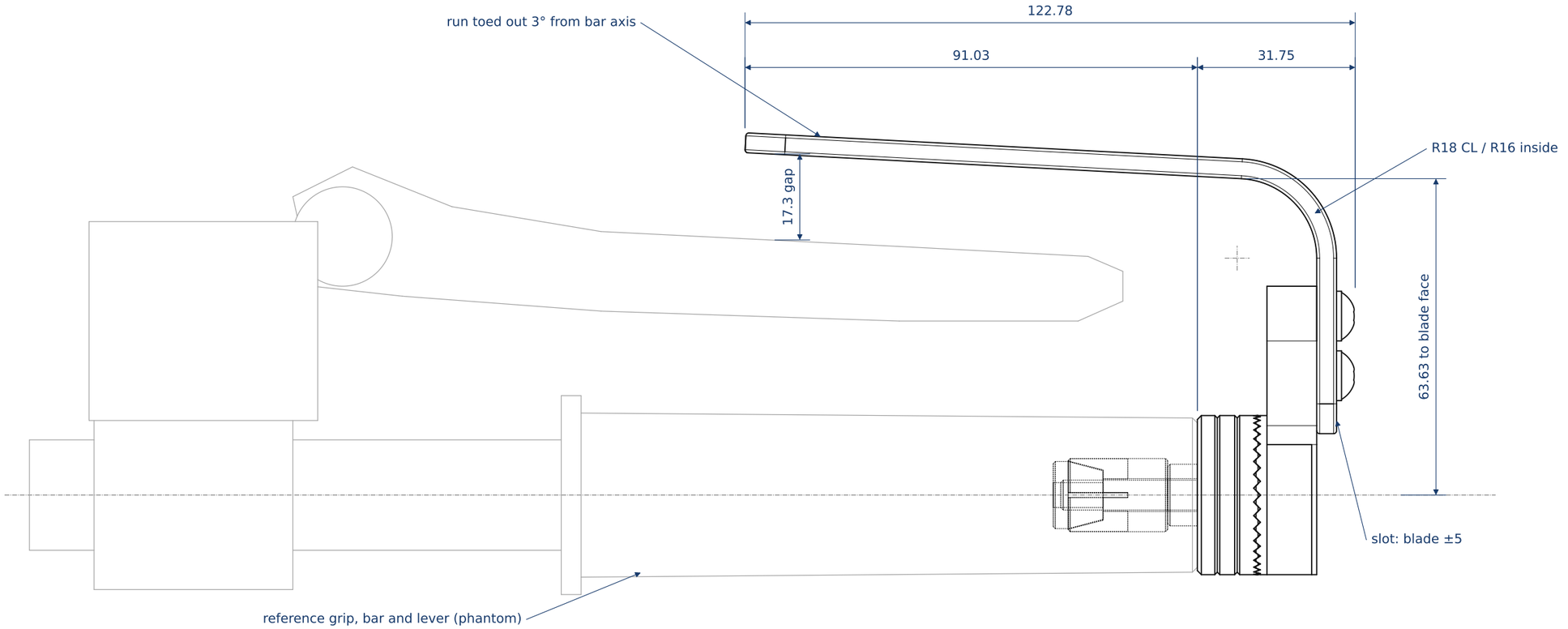
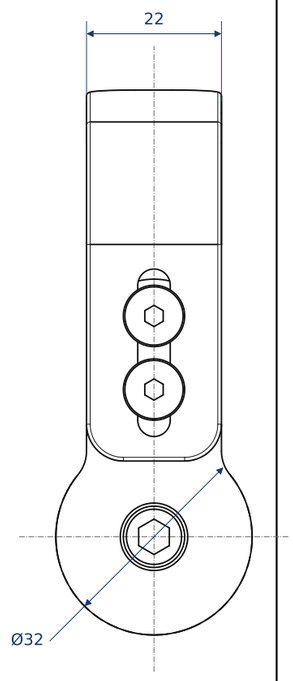
NOTES

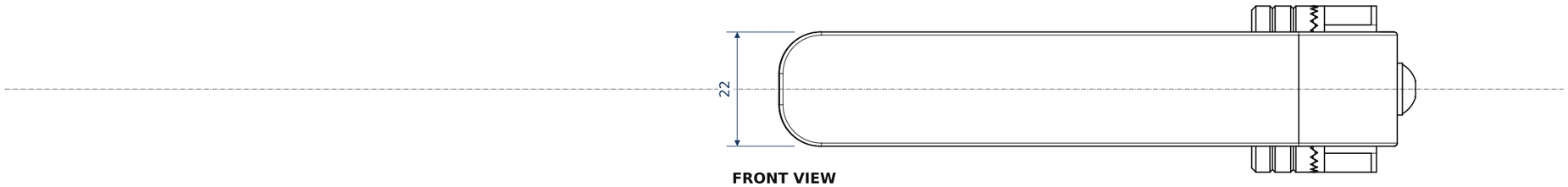
1. Drawing plane = lever plane. Turn the arm on its 36-tooth serration (10° steps) to put the blade on the lever line.
2. Clearances shown for the reference sport-bike lever; keep ≥ 10 mm all round, re-check at full lever travel.
3. Mass per side ≈ 109 g with the carbon blade and steel screws.
4. Left side: same parts, arm turned over (every part is symmetric about the lever plane).



PLAN (looking square onto the lever plane)



BAR-END VIEW



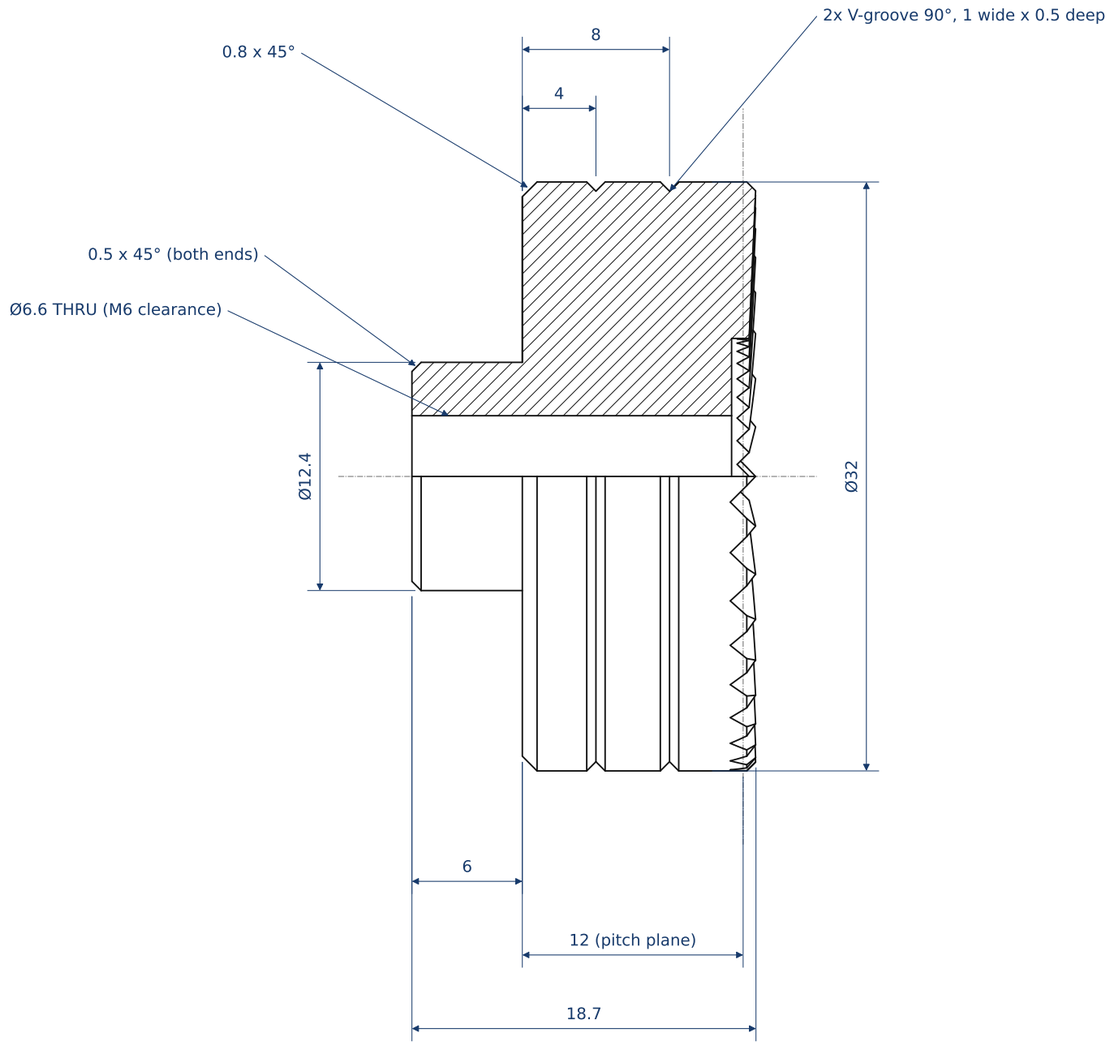
FRONT VIEW

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

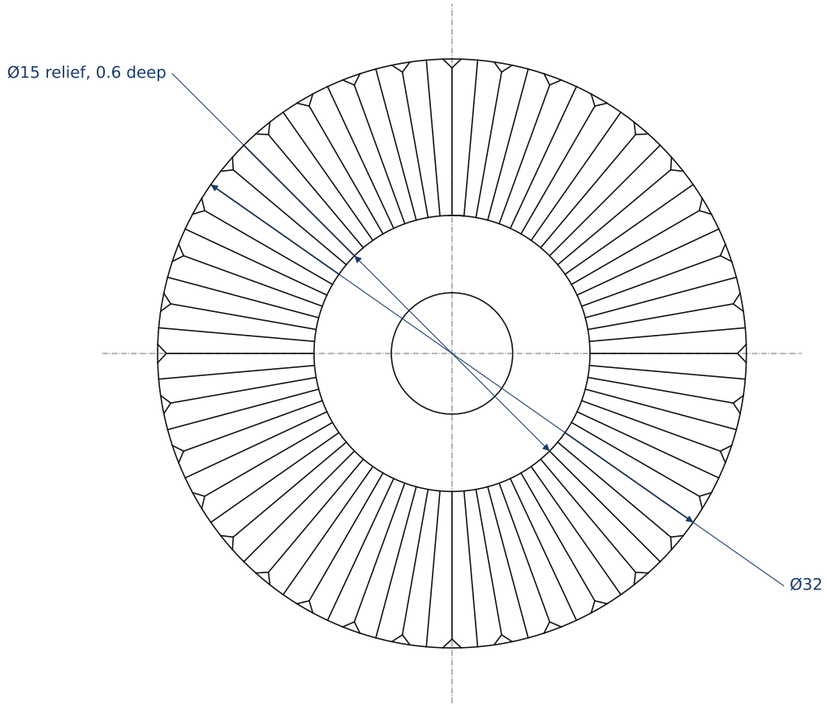
GENERAL ARRANGEMENT

LG-00

MATERIAL				FINISH	
see part sheets				see part sheets	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1 : 1	mm	1 set	ISO 2768-mK		1 / 5
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5



HALF SECTION A-A



VIEW ON SERRATION (from outboard)

NOTES

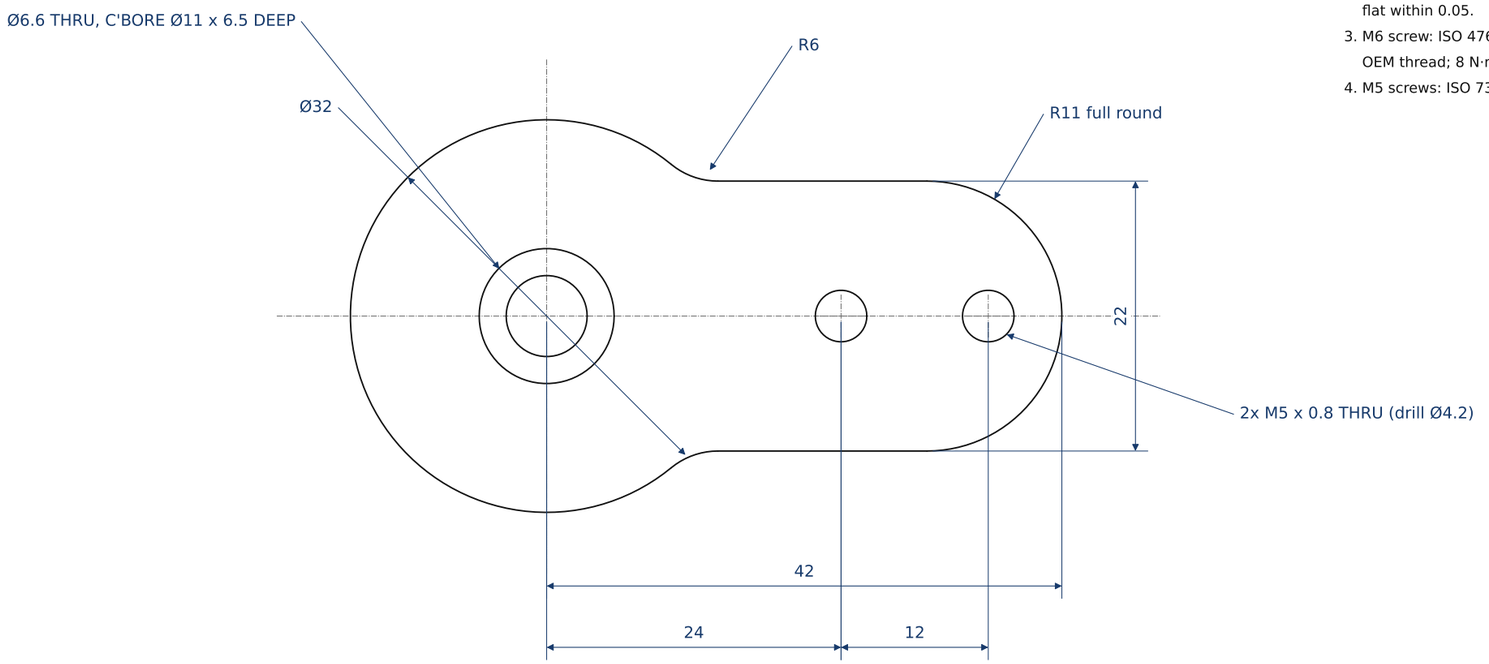
1. Hirth face serration: 36 teeth, 90° included, crests and roots radial through the axis, pitch plane at 12 from the bar-end face, toothed from Ø15 to Ø32: tooth height 1.4 at Ø32. Mates with LG-02 (offset half a tooth).
2. Variant LG-01b: same part without the spigot, for bars with a threaded OEM bar-end insert.
3. Spigot Ø12.4 centres in any bar with ID ≥ 12.5; the expander (LG-04/05) sits behind it.
4. Hard anodise after machining; mask nothing. Serration may be cut with a 90° V-cutter.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

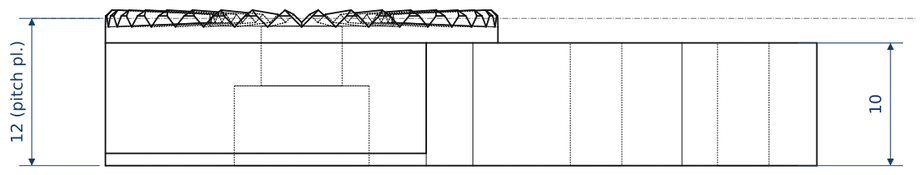
BAR-END ADAPTER

LG-01

MATERIAL			FINISH		
EN AW-6061-T6			Hard anodise, colour to order		
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
3 : 1	mm	1	ISO 2768-mK		2 / 5
SOURCE			DATE		BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py			2026-10-23		0.2 - 0.5



OUTBOARD FACE



PLAN (lever plane), serration toward the bar

NOTES

1. Hirth serration as LG-01: 36 teeth, 90°, pitch plane on the hub's inner face. Sits half a tooth out of phase with LG-01 so crests meet roots: 10° indexing.
2. Neck edges R1. Outboard face is the blade seat: flat within 0.05.
3. M6 screw: ISO 4762 M6 x 45 (expander) or to suit the OEM thread; 8 N·m, medium threadlocker.
4. M5 screws: ISO 7380 M5 x 12 + ISO 7089 washers, 4 N·m.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

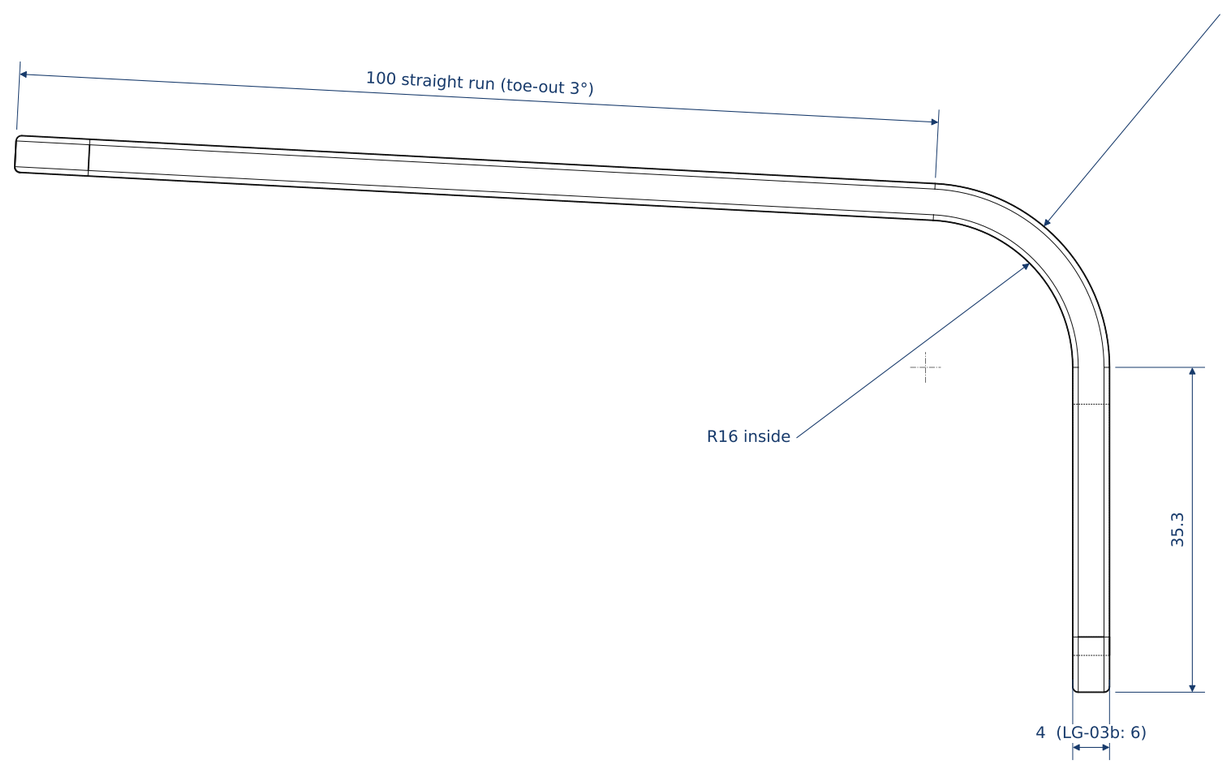
GUARD ARM

LG-02

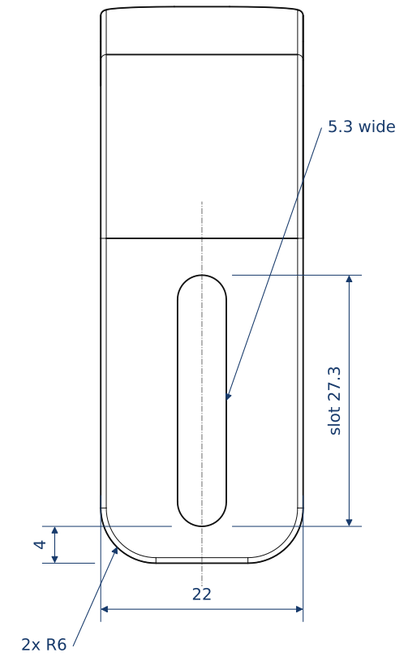
MATERIAL				FINISH	
EN AW-6061-T6				Hard anodise, colour to order	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
2 : 1	mm	1	ISO 2768-mK		3 / 5
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5

NOTES

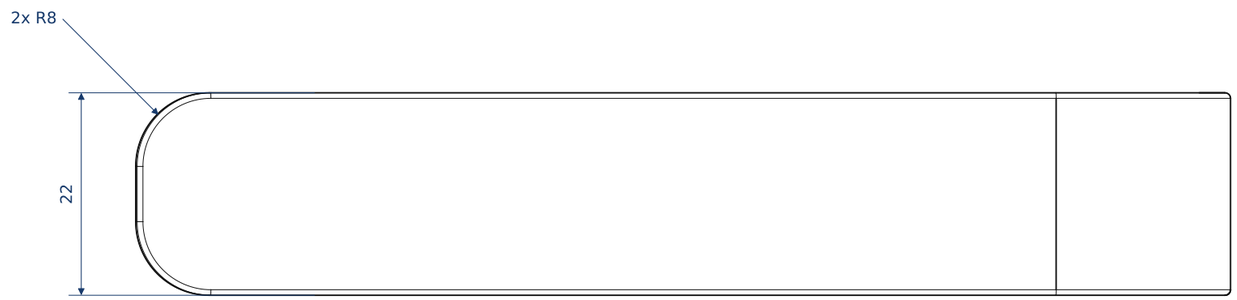
1. Centre-line length 162.6 mm: 35.3 straight + 87° bend on R18 + 100 straight.
2. LG-03 carbon: mould 4.0 ± 0.2 over a male tool (inside R16), cure, then CNC-trim outline and slot.
3. LG-03 aluminium option: bend 5052-H32 4 mm strip (blank 162.6 mm on the centre line), carbon hydro-dip.
4. LG-03b 3D print: same outline, 6 thick, PA12-CF / PETG-CF, print lying flat (lever plane on the bed),
   5 walls, 40 % gyroid; hydro-dip or wrap for the carbon finish.
5. Slot lets the blade slide ±5 mm toward / away from the lever.



PLAN (lever plane)



REAR PLATE (outboard)

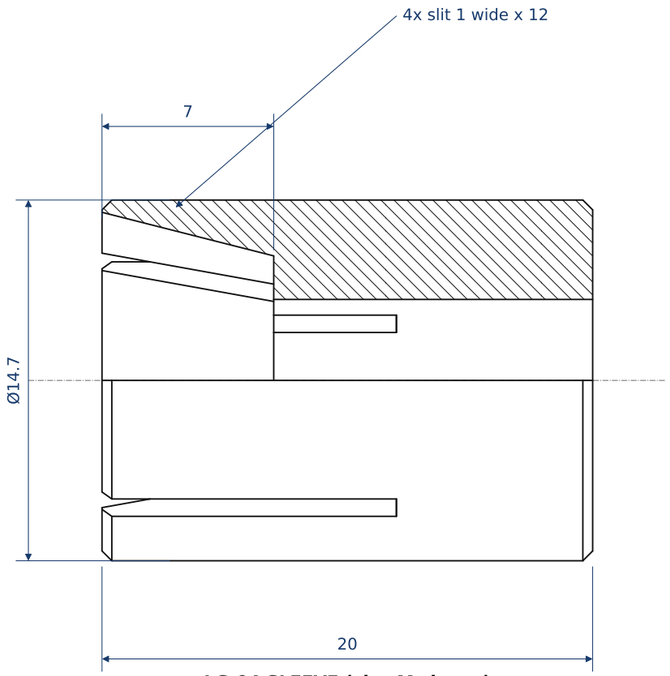


FRONT VIEW (outer face)

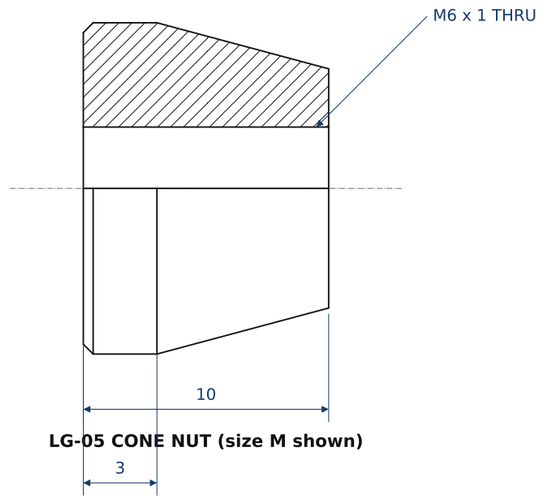
RACING LEVER GUARD / UNIVERSAL BAR-END FIT

GUARD BLADE — LG-03

MATERIAL: 3K 2x2 twill carbon / epoxy				FINISH: Gloss clear coat, edges R0.6	
SCALE: 3 : 2	UNITS: mm	QTY / SIDE: 1	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 4 / 5
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



LG-04 SLEEVE (size M shown)



LG-05 CONE NUT (size M shown)

SIZES (all lengths as drawn: sleeve 20, recess 7, cone land 3)

Size	Bar ID	Sleeve OD	Recess Ø face / bottom	Cone Ø big / small	Taper (half)
S	13 - 15	12.7	11.7 / 8.4	11.5 / 8	14°
M	15 - 17	14.7	13.7 / 10.15	13.5 / 9.75	15°
L	17 - 19	16.7	15.7 / 12.15	15.5 / 11.75	15°

NOTES

1. Bolt chain: M6 x 45 (ISO 4762) through LG-02 and LG-01 into the cone nut: 8.5 mm thread engagement at fit-up.
2. Tightening pulls the cone into the slit sleeve, which grips the bar bore. 8 N·m. Hold the arm square on its serration while tightening.
3. Pick the size from the bar's measured inside diameter.
4. Sleeve can be printed in TPU 95A (100 % infill) or cut from NBR tube; cone may be steel for higher load.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

EXPANDER SET

LG-04 / LG-05

MATERIAL: LG-04 TPU 95A or NBR 70A; LG-05 6061-T6				FINISH: LG-05 clear anodise or zinc	
SCALE: 4 : 1	UNITS: mm	QTY / SIDE: 1 sleeve + 1 cone	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 5 / 5
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5